Forward Backward Propagation

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梯度与雅各比矩阵

1.1 多元单值函数的梯度

定义 2.1. 设 $f(\boldsymbol{x}):\mathbb{R}^d\to\mathbb{R}$，定义梯度

$$\nabla f(\boldsymbol{x})=\left(\frac{\partial f}{\partial x_1},\ldots,\frac{\partial f}{\partial x_d}\right)^\top\in\mathbb{R}^{d\times 1} \tag{6}$$

1.2 多元多值函数的雅各比矩阵

定义 1.2. 设 $\boldsymbol{f}(\boldsymbol{x}):\mathbb{R}^d\to\mathbb{R}^p$，定义它的雅各比矩阵为：

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}=\begin{pmatrix}\frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{x_d}\\ \vdots & & \vdots\\ \frac{\partial f_p}{\partial x_1} & \cdots & \frac{\partial f_p}{\partial x_d}\end{pmatrix}_{p\times d} \tag{3}$$

梯度与雅各比矩阵

例 2.2. 设 $\boldsymbol{y} = W\boldsymbol{x}$, $\boldsymbol{x} \in \mathbb{R}^d$, $\boldsymbol{y} \in \mathbb{R}^p$, $W \in \mathbb{R}^{p\times d}$，求其雅各比矩阵。

将等式展开：

$$y_1 = w_{11}x_1 + w_{12}x_2 + \cdots + w_{1d}x_d \tag{8}$$

$$\cdots \tag{9}$$

$$y_p = w_{p1}x_1 + w_{p2}x_2 + \cdots + w_{pd}x_d \tag{10}$$

观察后不难发现

$$\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}} = W \tag{11}$$

链式法则

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1.3 链式法则的矩阵实现

设 $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x}) : \mathbb{R}^d \to \mathbb{R}^p, \quad f = f(\boldsymbol{y}) : \mathbb{R}^p \to \mathbb{R},$

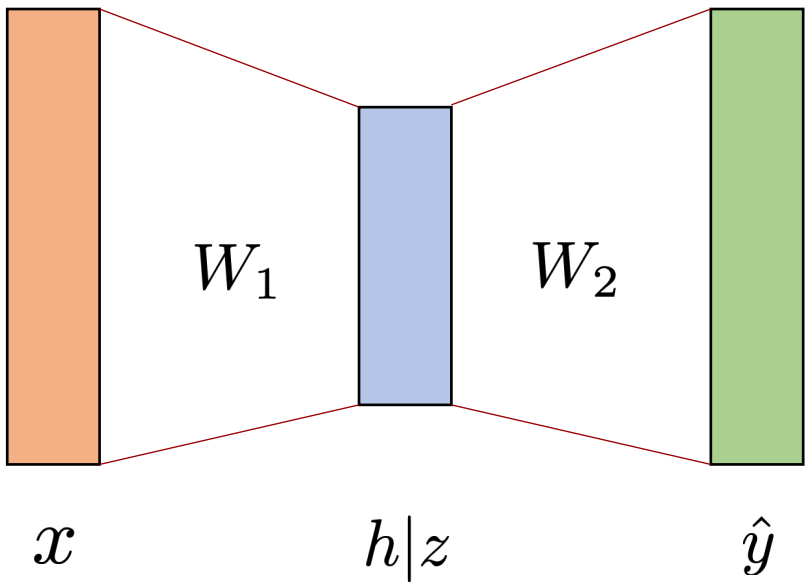
则有 $$\left(\frac{\partial f}{\partial x}\right)_{d\times 1} = \left[\left(\frac{\partial y}{\partial x}\right)^T\right]_{d\times p} \left(\frac{\partial f}{\partial y}\right)_{p\times 1}$$

重要引理

引理 2.6. 设 $f = f(y), \quad y = Wx, \quad x \in \mathbb{R}^d, \quad W \in \mathbb{R}^{p\times d}, \quad y \in \mathbb{R}^p$，则

$$\frac{\partial f}{\partial W} = \frac{\partial f}{\partial y} x^\top \tag{17}$$

两层神经网络



dims: d (p,d) p (q,p) q

$$h = W_1 x$$

$$z = \sigma(h)$$

$$\hat{y} = W_2 z$$

$$f = L(\hat{y})$$

Forward

$$\frac{\partial f}{\partial W_1} = \frac{\partial f}{\partial h} x^T$$

$$\frac{\partial f}{\partial h} = \frac{\partial f}{\partial z} \odot \nabla\sigma(h)$$

$$\frac{\partial f}{\partial W_2} = \frac{\partial L}{\partial \hat{y}} z^T, \quad \frac{\partial f}{\partial z} = W_2^{\mathsf{T}} \frac{\partial L}{\partial \hat{y}}$$

$$\frac{\partial f}{\partial \hat{y}} = \nabla L(\hat{y})$$

Backward

Store h, z and $\hat{y}$

Store $\nabla_{W_1} f(W_1)$ and $\nabla_{W_2} f(W_2)$